Improving CPU Efficiency for Data-Intensive Applications through Cache-Aware Spill Optimization

Register Spill Behaviour and Cache Hit/Miss Rates on RISC-V SPEC CPU2017

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gem5 TimingSimpleCPU · workload profile `test` · L1D 4 KiB, L1I 32 KiB, L2 32 KiB, L1D 2-way, L2 8-way · gem5 `e0a12aa4e3+dirty`

Superseded results (metric schema 8). The current detector (schema 9) also counts a reload whose address lies inside a wider or merged store (e.g. two 4-byte spills read by one 8-byte load). Schema 8 missed those, so `spill_load`, `spill_event` and the `other_stack` counts here are lower (up to 15% of the spill reloads in bwaves, 8.5% in leela, 3% in x264, under 0.1% in most benchmarks); instruction counts, spill stores and the cache behaviour of the pairs it found are unaffected.

Definitions (metric schema 8). Every stack store issued inside the region of interest, and every later load that reads it back (a load from the same address at the same width), belongs to exactly one class. `spill`: register-allocator spill code, identified by GCC itself (`-dP` RTL dumps of the same build, memory operands in the LRA spill area). `csave`: callee-saved register save/restore, return address included, identified from the binary's CFI (`.eh_frame`). `other_stack`: every remaining SP/FP-relative stack store and reload (stack-passed arguments, memory-resident locals). `stack_*` is the sum of the three. For each class an *event* is one (store, first reload) pair, so `event` $\leq$ `store` and `load` $\geq$ `event`. All counters cover only the region delimited by `m5_work_begin/m5_work_end`: the gem5 statistics are dumped and reset at those boundaries, so the cache columns span the same window, and the cache tags carry the class, so per-class hit/miss rates come straight from gem5. *Spill %* and *Csave %* are the class's store+reload instructions over `ROI_insts`; *spill/stack %* is the share of stack events that are real spills.

Benchmark	ROI_insts	stack_event	**spill_event**	csave_event	other_event	spill/stack %	Spill %	Csave %	L1D m %	L2 m %	sp_load m %	sp_store m %	cs_load m %	cs_store m %
cactuBSSN	48 023 254 960	3 817 969 294	3 794 919 348	22 956 427	93 519	99.4	21.32	0.10	15.764	64.32	4.455	3.914	0.126	0.159
parest	37 262 401 281	126 952 414	16 148 158	110 058 554	745 702	12.7	0.22	0.59	11.595	57.67	9.586	3.513	2.118	1.649
deepsjeng	46 202 686 169	1 436 088 677	380 604 124	1 046 384 737	9 099 816	26.5	2.78	4.53	9.980	6.89	2.859	7.613	3.282	3.898
leela	26 063 891 644	802 403 734	20 913 175	749 189 404	32 301 155	2.6	0.34	5.75	6.901	20.72	10.600	0.425	1.766	0.859
namd	31 914 751 099	336 698 329	334 454 030	1 939 340	304 959	99.3	3.13	0.01	12.887	25.71	5.486	1.313	3.844	3.891
lbm	6 506 800 040	274 131 044	274 130 500	544	0	100.0	9.67	0.00	23.735	88.75	0.424	2.261	15.074	0.184
nab	6 691 483 051	150 911 281	124 535 928	24 348 560	2 026 793	82.5	7.90	0.73	6.914	4.63	1.389	2.123	0.898	1.211
xz	2 014 351 521	32 945 274	4 682 897	26 392 608	1 869 769	14.2	0.65	2.62	6.496	25.90	0.133	1.002	2.222	2.241
xz_4m_l1	2 001 462 938	29 215 304	3 958 947	23 548 312	1 708 045	13.6	0.57	2.35	7.996	35.15	0.131	0.808	2.863	1.815
xz_4m_l2	2 083 103 943	28 125 372	3 700 138	22 776 791	1 648 443	13.2	0.51	2.19	9.797	42.07	0.126	0.814	3.396	1.908
xz_4m_l3e	10 573 516 663	124 670 474	54 899 409	66 069 008	3 702 057	44.0	2.44	1.25	6.916	19.19	2.000	1.167	1.747	3.242
xz_4m_l4	4 011 967 585	66 558 242	20 142 751	44 411 548	2 003 943	30.3	2.11	2.21	8.369	30.25	2.282	2.394	1.558	2.249
xz_4m_l4e	12 115 711 352	127 720 342	56 348 941	67 532 844	3 838 557	44.1	2.22	1.11	6.702	17.71	2.107	1.170	1.810	3.475
xz_1m_l0	711 271 503	12 737 212	1 091 050	10 938 543	707 619	8.6	0.56	3.08	5.939	29.24	0.192	2.012	1.021	1.382
xz_1m_l1	719 848 327	12 124 762	973 253	10 458 508	693 001	8.0	0.52	2.91	6.727	33.78	0.193	1.706	1.215	1.441
xz_1m_l2	724 690 222	11 886 991	956 147	10 241 945	688 899	8.0	0.51	2.83	6.928	34.23	0.170	1.718	1.254	1.484
xz_1m_l3e	1 994 406 562	32 343 460	11 257 805	20 010 020	1 075 635	34.8	2.49	2.01	6.214	18.66	1.647	2.131	0.775	2.798
xz_1m_l4	1 076 057 785	22 448 482	5 149 009	16 486 476	812 997	22.9	1.97	3.06	6.832	24.25	1.789	4.355	0.697	2.351
xz_1m_l4e	2 198 787 666	32 786 002	11 469 277	20 222 586	1 094 139	35.0	2.33	1.84	6.173	17.52	1.725	2.109	0.800	2.919
imagick	117 733 663	2 444 391	2 450	2 437 084	4 857	0.1	0.02	4.14	4.422	81.36	10.072	69.568	1.301	0.348
blender	994 421 804	46 392 827	4 430 041	40 844 530	1 118 256	9.5	1.16	8.21	5.341	19.43	2.281	3.165	2.206	2.767
povray	2 240 986 547	168 771 227	2 121 674	155 851 654	10 797 899	1.3	0.27	13.91	10.145	41.21	7.406	20.677	1.842	1.807
omnetpp	12 426 241 547	842 378 057	5 233 032	791 024 854	46 120 171	0.6	0.09	12.73	12.566	26.58	0.250	0.859	1.929	2.480
xalancbmk	395 919 938	18 465 753	295 456	17 282 226	888 071	1.6	0.23	8.73	13.661	33.74	10.003	18.696	3.437	4.325
x264	475 535 613 470	19 188 657 460	13 318 402 468	5 322 257 634	547 997 358	69.4	6.15	2.24	2.326	26.97	0.399	0.954	1.487	1.610
gcc	16 026 870	1 020 580	52 732	952 288	15 560	5.2	0.84	11.88	7.123	32.69	0.717	1.875	1.331	1.231
mcf	24 426 982 047	234 360 896	91 410 910	142 147 076	802 910	39.0	1.58	1.16	25.499	75.88	5.567	8.282	3.676	5.825
perlbench_makerand	2 421 667 618	224 582 067	55	224 562 979	19 033	0.0	0.08	18.55	5.676	0.00	0.000	2.725	1.717	0.873
bwaves_1	17 288 358 363	145 483 579	145 451 376	26 481	5 722	100.0	3.55	0.00	41.067	31.22	2.704	0.330	4.063	5.189
bwaves_2	17 287 430 556	145 483 497	145 451 376	26 421	5 700	100.0	3.55	0.00	41.069	31.22	2.704	0.330	4.069	5.193
exchange2	57 667 143 431	1 874 700 399	1 304 759 851	345 493 349	224 447 199	69.6	5.03	1.20	2.504	1.08	2.744	3.560	7.229	4.833
fotonik3d	3 098 360 920	28 393 792	27 853 217	437 206	103 369	98.1	3.84	0.03	17.861	41.35	5.709	0.057	1.160	3.747
roms	28 955 766 090	203 000 119	182 774 988	12 238 510	7 986 621	90.0	4.79	0.08	20.998	38.82	4.437	1.564	1.413	2.838
wrf	84 396 898 728	420 492 655	109 777 107	285 402 467	25 313 081	26.1	0.91	0.68	18.055	22.28	9.397	6.610	0.844	1.636
cam4	33 466 803 539	584 634 561	168 239 124	404 426 599	11 968 838	28.8	2.11	2.42	10.243	52.99	2.650	2.913	1.165	1.702
Total	1 003 626 799 442	31 607 978 550	**20 626 590 744**	10 039 378 113	942 009 693	65.3	5.12	2.00	0.000	—	—	—	—	—